

Exercise: LangChain Memory & HITL

1. Objective

Develop a comprehensive AI-powered Travel Consultant agent. The goal is to build an intelligent assistant capable of maintaining context across conversations, remembering user preferences over long periods, staying within strict conversational boundaries, requesting human approval for sensitive actions, and being fully observable for quality assurance.

2. Context and Short-Term Memory Setup

Establish the foundation of the conversational agent.

- **System Prompt & Context:** Define a strict persona for the agent. It should act as an exclusive, high-end travel consultant. Inject context about the agency's premium services, available luxury destinations, and standard booking procedures.
- **Short-Term Memory Integration:** Implement a conversational memory mechanism. The agent must be able to maintain the context of the immediate conversation. During the interaction, the user will specify multiple requirements across different prompts (e.g., choosing a destination, then selecting a hotel, then picking flight dates). The agent must remember all these details seamlessly within the current active session without the user repeating them.

3. Long-Term Memory Implementation

Enable the agent to provide a highly personalized experience by remembering details across multiple separate sessions.

- **Preference Extraction:** Design the agent to identify and extract persistent user preferences from the chat (e.g., dietary restrictions, seating preferences on flights, or favorite hotel brands).
- **Persistent Storage Integration:** Implement a mechanism to store these extracted facts. When a new session begins, the agent must retrieve this historical data and proactively apply it to the new conversation, proving that it "remembers" the user from previous interactions.

4. Implementing Guardrails

Ensure the agent remains professional, safe, and aligned with business goals.

- **Topical Boundaries:** Implement strict conversational guardrails. The agent must politely decline to answer any questions unrelated to luxury travel. It must specifically block and redirect inquiries about budget travel, competitors' agencies, and controversial topics like politics.
- **Output Formatting Guardrails:** Ensure that every time the agent presents a price or a final itinerary, the output strictly adheres to a predefined structured format, preventing hallucinations in pricing or service descriptions.

5. Human-in-the-Loop (HITL) Integration

Create a secure workflow for high-stakes actions.

- **Action Interruption:** Set up a tool or function for "Finalizing the Booking and Processing Payment."

- **Approval Workflow:** Configure the workflow so that the agent cannot execute this final booking tool autonomously. Instead, the process must pause and request explicit human authorization. The system should present a summary of the action to be taken, wait for human approval (or rejection), and then resume the conversation based on the human's decision.

6. Observability and Evaluation (LangSmith)

Implement tracing and quality control to monitor the agent's behavior in real-time.

- **Full Tracing:** Connect the application to the observability platform. Ensure that every user input, agent thought process, tool execution, and memory retrieval step is fully logged and traceable.
- **Evaluation Metrics:** Create an evaluation pipeline. Define specific criteria to grade the agent's responses. The criteria should evaluate whether the agent successfully applied long-term memory, maintained its high-end persona, and successfully triggered the guardrails when provoked.